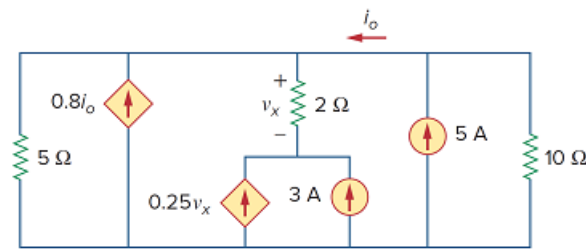


Problem

Find v_x and i_x in the circuit shown in Fig. 3.107.

Figure 3.107:



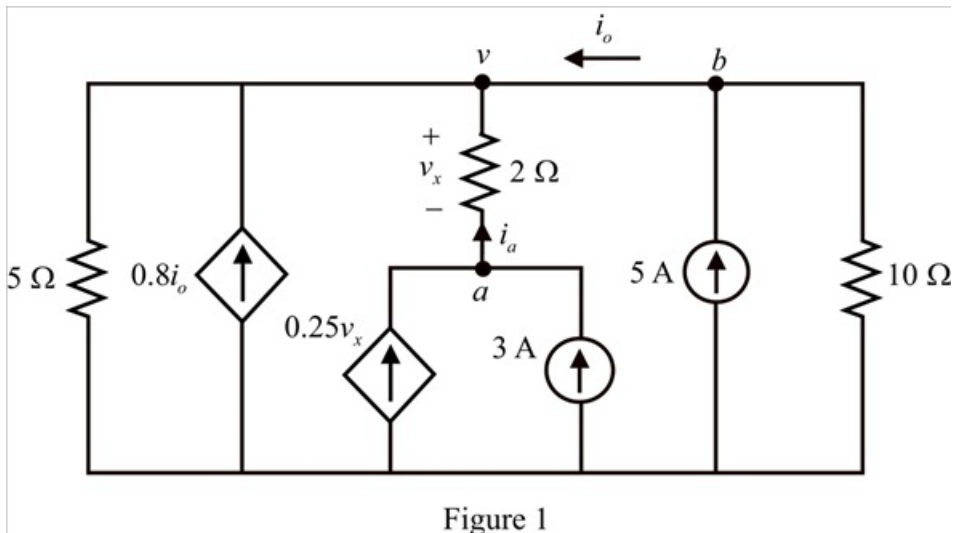
Step-by-step solution

Step 1 of 5

Refer to the circuit shown in Figure 3.107 in the textbook.

Note that there is no i_x marked the circuit, so consider it as a typo error and take it as i_o .

Draw the circuit diagram with node notations.



Step 2 of 5

Apply Kirchhoff's current law at node a .

$$i_a = 0.25v_x + 3$$

Write the expression for the voltage v_x using Ohms law.

$$v_x = 2(-i_a)$$

$$v_x = -2(0.25v_x + 3)$$

$$v_x = -0.5v_x - 6$$

$$v_x + 0.5v_x = -6$$

$$1.5v_x = -6$$

$$v_x = -\frac{6}{1.5}$$
$$= -4 \text{ V}$$

Therefore, the value of the voltage v_x is $\boxed{-4 \text{ V}}$.

Step 3 of 5

Assume the voltage at the common node to be v .

Apply Kirchhoff's current law at node b .

$$i_o + \frac{v}{10} = 5$$

$$\frac{v}{10} = 5 - i_o$$

$$v = 50 - 10i_o$$

Therefore, the expression for the voltage v is $50 - 10i_o$.

Step 4 of 5

Apply Kirchhoff's current law at common node, v .

$$\frac{v}{5} - 0.8i_o - i_a - 5 + \frac{v}{10} = 0$$

$$\frac{v}{5} - 0.8i_o - 0.25v_x - 3 - 5 + \frac{v}{10} = 0 \quad \text{since } i_a = 0.25v_x + 3$$

$$\frac{3v}{10} - 0.8i_o - 0.25v_x - 8 = 0$$

Step 5 of 5

Substitute $50 - 10i_o$ for v and -4 V for v_x .

$$\frac{3(50 - 10i_o)}{10} - 0.8i_o - 0.25(-4) - 8 = 0$$

$$\frac{150 - 30i_o}{10} - 0.8i_o + 1 - 8 = 0$$

$$15 - 3i_o - 0.8i_o - 7 = 0$$

$$-3.8i_o + 8 = 0$$

$$3.8i_o = 8$$

$$i_o = \frac{8}{3.8}$$

$$i_o = 2.105 \text{ A}$$

Therefore, the value of the current i_o is $\boxed{2.105 \text{ A}}$.